CYBER.SOHO

Educational Hub

From the Industry to the Classroom — Building the IT's Future, Today and Together!

DAY 04

IPv4 & IPv6

Sub- and Super-netting

Course details

- Course: Basics of Computer Networks (24 h, 6 sessions)
- Reference: Cisco NetAcad — Networking Essentials 3.0
- Session: Day 04 — Thursday, May 7, 2026 (4 h)
- Project Stage 4: Address the Network

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Today's Agenda — 4 blocks, 1 hour each

01

LECTURE 1

The IPv4 Address

binary ↔ decimal • subnet masks • classes A–E • CIDR / VLSM • RFC 1918 • casts • subnetting & supernetting

02

EXERCISE 1

Subnetting Workshop

Build personal cheat-sheet • magic-number method • 20 problems across 5 blocks • online verification

03

LECTURE 2

Major Related Protocols

static vs dynamic • DHCP / DORA • NAT / PAT • IPv6 — why & why not yet

04

EXERCISE 2

Project Stage 4 — Address the Network

Acme SoHo IP plan • GNS3 router DHCP • IPFire on VirtualBox • pfSense VLANs + DMZ + DHCP relay

BLOCK 01

Lecture 1

The IPv4 Address — 1 hour

1.1 Binary ↔ Decimal Conversion

The 32-bit IPv4 address

- 32 bits split into 4 octets of 8 bits each
- Written as dotted decimal: 192.168.1.1
- Each octet → decimal value 0–255
- Routers speak binary; decimal is just lipstick
- You MUST be able to convert in your head

32 bits = 4 octets × 8 bits

11000000	10101000	00000001	00000001
192	168	1	1

→ **192.168.1.1**

🎯 **KEY TAKEAWAY:** *The 8-bit weights 128 64 32 16 8 4 2 1 are the most useful row of numbers in your career — every IPv4 problem starts there.*

1.1 Binary ↔ Decimal — The cheat sheet

Memorise this row — it never leaves you

weight	128	64	32	16	8	4	2	1
for 200	1	1	0	0	1	0	0	0

Example — convert 200 to binary (greedy subtraction):

- 128 fits → write 1 (remainder 72) → 64 fits → write 1 (remainder 8)
- 32 too big → 0 • 16 too big → 0 • 8 fits → 1 (remainder 0)
- 4 too big → 0 • 2 too big → 0 • 1 too big → 0 ⇒ 11001000 ✓

🎯 **KEY TAKEAWAY:** *Binary → Decimal: sum the weights where there is a 1. Decimal → Binary: greedy subtraction starting at 128.*

1.2 Subnet Masks

The line between neighbourhood and house number

- 32-bit value: all 1s on the left (network), all 0s on the right (host)
- Two equivalent notations:
  - Dotted-decimal: 255.255.255.0
  - CIDR slash: /24 (= 24 ones)
- Reserved addresses inside every subnet:
  - First address → network identifier (cannot be assigned)
  - Last address → broadcast (cannot be assigned)
- Usable hosts = total addresses − 2

Example — 192.168.1.0/24

NETWORK (24 bits)	HOST (8 bits)

192.168.1.0

255.255.255.0 = /24

Inside this /24:

- Network: 192.168.1.0
- Broadcast: 192.168.1.255
- Usable: .1 → .254
- Hosts: 254

🎯 **KEY TAKEAWAY:** *Get the mask wrong and packets walk into the wrong building.*

1.2 Subnet Masks — Common values

Print this; tape to your monitor

CIDR	Subnet Mask	Block size	Total addresses	Usable hosts
/24	255.255.255.0	256	256	254
/25	255.255.255.128	128	128	126
/26	255.255.255.192	64	64	62
/27	255.255.255.224	32	32	30
/28	255.255.255.240	16	16	14
/29	255.255.255.248	8	8	6
/30	255.255.255.252	4	4	2

🎯 **KEY TAKEAWAY:** *Memorise /24, /27, /30. The rest you can derive from the cheat-sheet in seconds.*

1.3 Classes A, B, C, D, E (historical)

Pre-CIDR address allocation — RFC 791, 1981

- Determined the default mask from the first octet alone
- Wasted huge address blocks → replaced by CIDR in 1993
- Still alive in OS defaults, exam questions, and old documentation
- Special: first octet 0 = 'this network', 127 = loopback (127.0.0.1)

Class	First octet	Default mask	Purpose	Madrid analogy
A	1–126	/8	Huge networks	*Real Madrid*
B	128–191	/16	Mid-size networks	*Primera División*
C	192–223	/24	Small networks	*Sunday-league club*
D	224–239	n/a	Multicast	*Bernabéu PA system*
E	240–255	rsvd	Experimental	*Locked storage room*

🎯 **KEY TAKEAWAY:** *Classful is dead in routing, but alive in OS defaults. Recognise on sight; never rely on it for design.*

1.4 Classless / CIDR / VLSM

RFC 1519 (1993) — the day the Internet stopped wasting addresses

CIDR — Classless Inter-Domain Routing

- Drops rigid class boundaries
- Mask is whatever you say — /13, /27, /30
- Notation: address / prefix length
- Example: 192.168.1.0/24
- Enabled the modern Internet (route aggregation)

VLSM — Variable Length Subnet Masking

- Different subnets can have different mask sizes
- Fits subnet size to actual host count (no waste)
- Allocation rule: ALWAYS carve largest first
- Daily reality of every IP plan you write

🎯 **KEY TAKEAWAY:** *Classful = pre-built suit. VLSM = tailored suit. The latter fits, the former wastes fabric.*

1.5 Public vs Private — RFC 1918

The three private ranges every IT person knows by heart

10.0.0.0/8	172.16.0.0/12	192.168.0.0/16
16,777,216 addresses	*1,048,576 addresses*	*65,536 addresses*
Class A scale Most common in large enterprises	Class B scale Middle range — often forgotten	Class C scale Every home router on Earth

Properties of private addresses

- Reusable across organisations — millions of LANs use 192.168.1.0/24
- NOT routable on the public Internet
- Must be NATted (Network Address Translation) to talk outward — see Lecture 2.3

🎯 **KEY TAKEAWAY:** *Same apartment number '1A' exists in 50 buildings. The street address (public IP) is what makes mail get delivered from outside.*

1.5 Special-purpose ranges (not 1918, but not routable)

Recognise these on sight in Wireshark

`127.0.0.0/8`	Loopback (talk to yourself)	*RFC 1122*
`169.254.0.0/16`	APIPA — DHCP failed fallback	*RFC 3927*
`100.64.0.0/10`	CGNAT — your mobile carrier	*RFC 6598*
`224.0.0.0/4`	Multicast (Class D)	*RFC 5771*
`192.0.2.0/24`	TEST-NET-1 (documentation only)	*RFC 5737*
`198.51.100.0/24`	TEST-NET-2 (documentation only)	*RFC 5737*
`203.0.113.0/24`	TEST-NET-3 (documentation only)	*RFC 5737*
`255.255.255.255/32`	Limited broadcast (DHCP Discover)	*RFC 919*

🎯 **KEY TAKEAWAY:** *If you see 169.254.x.x on a NIC: DHCP is broken. If you see 100.64.x.x: you're behind ISP CGNAT.*

1.6 Unicast / Multicast / Broadcast

Three ways a packet can be addressed

UNICAST

ONE → ONE

- Default for almost all traffic
- Source IP → single destination IP
- Example: your laptop opens google.com

"Order one croqueta from one waiter"

MULTICAST

ONE → SUBSCRIBED GROUP

- Class D range 224.0.0.0/4
- Receivers must subscribe (IGMP)
- Examples: OSPF (224.0.0.5), mDNS, IPTV

"Waiter shouts the football score — only the table that subscribed cares"

BROADCAST

ONE → EVERYBODY

- Limited: 255.255.255.255 (LAN-only)
- Directed: all-ones host of any subnet
- Used by DHCP Discover, ARP requests

"Bar owner: 'we close in 5 minutes!'"

🎯 **KEY TAKEAWAY:** *Switches FLOOD broadcasts; routers BLOCK them. VLANs and subnetting CONTAIN broadcast domains.*

1.7 Subnetting & Supernetting

Same arithmetic, opposite directions

⬇ SUBNETTING

Split a big block into many small ones

- Why: broadcast control, security, audit
- How: borrow host bits → more network bits
- +1 to prefix → halves size, doubles count
- Inside the building

⬆ SUPERNETTING

Merge contiguous blocks into one

- Also called: route summarisation
- Why: smaller routing tables, faster lookups
- Inputs must be contiguous AND aligned
- At the lobby directory

Example → four contiguous /24s aggregate to one /22

`10.42.0.0/24` `10.42.1.0/24` `10.42.2.0/24` `10.42.3.0/24` → `10.42.0.0/22`

Lecture 1 — In one minute

What you must own before lunch

1. Convert any decimal 0–255 to/from binary using the row 128 64 32 16 8 4 2 1
2. Read any /N mask and quote: network, broadcast, first/last host, host count
3. Recognise classes A/B/C on sight from the first octet — but never rely on them for design
4. CIDR + VLSM = the modern way; allocate LARGEST FIRST always
5. Three RFC 1918 ranges in your sleep: 10/8, 172.16/12, 192.168/16
6. Identify unicast / multicast / broadcast from the destination IP alone
7. Subnet inside the building, supernet at the lobby directory

BLOCK 02

Exercise 1

Subnetting Workshop — 1 hour

Exercise 1 — Subnetting Workshop (1 h)

Build a personal cheat-sheet, then solve 20 problems

1

personal cheat-sheet

Powers of 2, /24 → /30 table, magic-number method

5

blocks of problems

Network/broadcast • Host counts • VLSM • Supernetting • Mixed

20

problems total

Hand-solve, then verify with online calculator screenshots

🎯 **KEY TAKEAWAY:** *First 5 problems will hurt. Problems 15–20 will feel automatic. That's the workshop's whole point.*

Exercise 1 — The Magic-Number Method

Five steps, no calculator, repeatable

1

Find the interesting octet

Where the mask transitions from 255 to <255

2

Compute block size

256 − value of that mask octet

3

Find boundaries

Multiples of block size = subnet boundaries

4

Network & broadcast

Network = lower • Broadcast = next−1

5

First & last host

Network+1 and Broadcast−1

Worked example — 172.16.50.130/27

- /27 mask = 255.255.255.224 → interesting octet is the 4th (224)
- Block size = 256 − 224 = 32 → boundaries: 0, 32, 64, 96, 128, 160, 192, 224
- 130 lands between 128 and 160 → network 172.16.50.128, broadcast 172.16.50.159, hosts .129–.158

Exercise 1 — The 20 problems

Five blocks of four — submit with calculator screenshots

BLOCK A	BLOCK B	BLOCK C	BLOCK D	BLOCK E
Network & Broadcast	***Subnet & Host Counting***	***VLSM Planning***	***Supernetting***	***Mixed Classifier***
192.168.5.45 /24	10.0.0.0/22	10.42.0.0/22 → /25 + /26 + /27 + /28 + /29	192.168.4.0/24 ... .7.0/24	169.254.10.20 — what kind?
10.20.30.40 /20	192.168.40.0/26	192.168.0.0/24 → 50+25+10+2 hosts	10.0.0.0/24 ... .7.0/24	100.64.5.5 — what kind?
172.16.50.130 /27	172.20.0.0/19	172.16.0.0/20 → 4 × 500 hosts	172.16.0.0/24 + 172.16.2.0/24 ?	224.0.0.5 vs 192.168.1.0/24 — cast?
192.168.100.99 /29	198.51.100.0/30	10.10.0.0/16 → /22 + 2×/23 + 4×/24	192.168.32.0/24 ... .35.0/24	192.168.1.255 vs /24 — cast?

Exercise 1 — Deliverables

What lands in the LMS by next week

1-page personal cheat-sheet

PDF or photo — must fit on one A5

All 20 answers solved by hand

Show the magic-number method

Calculator screenshots

Verify each answer with subnet-calculator.com or jodies.de/ipcalc

Method paragraph per block

One paragraph explaining your approach

Filename

Day04_Exercise1_<your_name>.pdf

BLOCK 03

Lecture 2

Major Related Protocols — 1 hour

2.1 Static vs Dynamic IP Addressing

Two ways to put an IP on a NIC

STATIC

Manually configured

- Reliable; never changes
- High maintenance — every move = config edit
- USE FOR: servers, printers, switches/routers
- Anything that DNS or firewalls reference

DYNAMIC

Assigned by DHCP

- Hands-off; one config (the DHCP server)
- Lease-based; addresses recycle automatically
- USE FOR: workstations, laptops, phones
- BYOD, guest Wi-Fi

HYBRID

DHCP reservations

- Best of both worlds
- Same IP every time, central management
- Server matches on MAC address
- USE FOR: printers, IP cameras, IoT

🎯 **KEY TAKEAWAY:** *Servers static, clients dynamic, printers reserved. Memorise the rule and 90% of 'should this be static?' goes away.*

2.2 DHCP — DORA Process

The four-message dance every IT person knows by heart

D ▶ **O** ▶ **R** ▶ **A**

DISCOVER

Client broadcasts:
'any DHCP server out there?'

→ `255.255.255.255`

OFFER

Server(s) reply with:
IP + mask + GW + DNS + lease

← `unicast or broadcast`

REQUEST

Client picks one offer,
broadcasts the choice

→ `others withdraw`

ACK

Chosen server commits
the lease and confirms

← `binding active`

🎯 **KEY TAKEAWAY:** *DORA is THE network admin's daily Wireshark filter. If you can't quote it cold, you're not done with this lecture.*

2.2 DHCP Options — what gets handed to the client

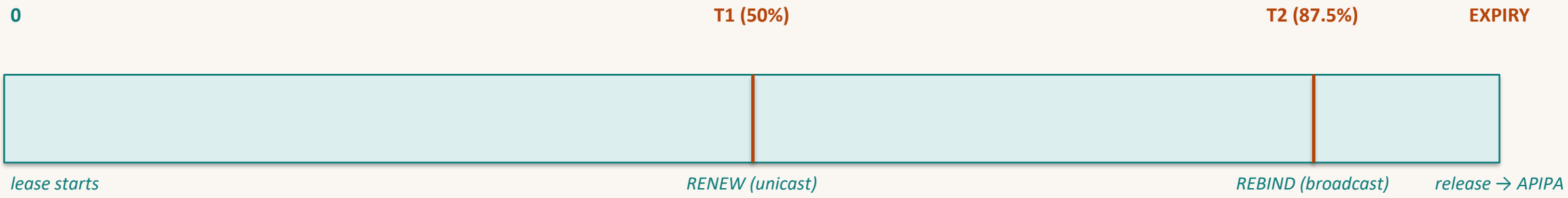
RFC 2131 + RFC 2132

Opt #	Name	Example value	Why it matters
1	Subnet mask	255.255.255.0	Tells client where the LAN ends
3	Default gateway	192.168.1.1	First hop for everything off-subnet
6	DNS servers	1.1.1.1, 8.8.8.8	Name resolution
15	Domain name	corp.acme.local	DNS suffix for short names
42	NTP servers	pool.ntp.org	Clock sync — vital for logs & TLS
51	Lease duration	8 hours, 24 hours, ...	When client must renew
66	TFTP server	10.0.0.50	PXE boot, IP-phone provisioning
82	Relay agent info	circuit-id, port-id	RFC 3046 — for security & accounting

🎯 **KEY TAKEAWAY:** *Knowing options 1, 3, 6, 51 by heart will solve 90% of 'why can't this device get on the network?' tickets.*

2.2 Lease lifecycle — the timers

Why your laptop usually keeps the same IP



'Short DORA' — the unicast renewal

- At T1, client doesn't broadcast — it sends a UNICAST REQUEST direct to the original server
- Server responds with ACK; lease timer reset
- Just R + A (Request + Acknowledge) — much less network noise than full DORA
- If T2 hits without renewal, client broadcasts to find ANY server (REBIND)

2.2 DHCP Relay & Server Redundancy

Crossing routers and surviving server failure

DHCP Relay (ip helper-address)

- DHCP is broadcast; broadcasts don't cross routers
- Relay agent on the router intercepts Discover/Request
- Forwards them as UNICAST to a remote DHCP server
- Required when DHCP server is on different VLAN
- Cisco IOS: ip helper-address <server-ip>
- Adds RFC 3046 Option 82 (circuit-id, port-id)
- Used for security, accounting, anti-spoofing

Two-server redundancy

Failover (RFC 8156, ISC Kea, Microsoft)

- Servers split scope and SYNC state
- If one fails, other takes over both halves
- Modern, recommended default

Split-scope (classic 80/20)

- Servers DON'T sync — each owns part
- Simple but allows brief duplicate offers
- Still common in legacy ISC dhcpd setups

2.3 NAT / PAT — The problem and its flavours

Why we still have IPv4 in 2026

The problem NAT solves

- Only ~4.3 billion IPv4 addresses exist (2^{32}) — far more devices than that
- Solution: many private hosts share a few (or one) public IP via translation

Static NAT

`1 : 1 fixed`

One private ↔ one public, fixed mapping. Used when an internal server needs a stable public face.

Dynamic NAT

`Pool : Pool`

Pool of public IPs, first-come-first-served. Once free pool is empty, new internal hosts wait.

PAT (NAT-overload)

`Many : 1 + ports`

Many private → one public, separated by source port. Your home router. The default everywhere.

🎯 **KEY TAKEAWAY:** *PAT is the default. Static NAT is for servers. Dynamic NAT is mostly historical.*

2.3 PAT — How the translation table works

The Hotel Ritz switchboard, in code

Public IP: 203.0.113.50 • Sample translation table:

Inside (priv)	Outside (public)	Destination	Proto
10.0.0.10:51234	203.0.113.50:50032	8.8.8.8:53	UDP
10.0.0.11:50111	203.0.113.50:50033	142.250.74.78:443	TCP
10.0.0.12:49500	203.0.113.50:50034	151.101.1.69:443	TCP
10.0.0.10:51235	203.0.113.50:50035	142.250.74.78:443	TCP
10.0.0.11:50112	203.0.113.50:50036	1.1.1.1:53	UDP

Limitations & pain points

- Inbound unsolicited connections — broken by default; need port forwarding
- Some protocols carry IPs in payload (SIP, FTP active, IPsec) — need ALGs
- Logging is required for forensics — and for GDPR (you must know who did what, when)

2.4 IPv6 — Why and Why Not Yet

The future has been 'next year' for 25 years

Why IPv6 will replace IPv4

- Address exhaustion: IANA's pool ran out 3 Feb 2011
- 128-bit space: 2^{128} addresses (≈340 undecillion)
- Simpler header — fewer fields, no checksum
- Built-in autoconfig — SLAAC, no DHCP needed
- Native multicast & link-local
- End of NAT — true end-to-end addressing
- Mandatory for IoT scale

Why not yet (the migration drag)

- Massive installed base of v4-only kit & apps
- IT staff trained in v4 — re-training is real cost
- Management tools may not support v6 fully
- NAT works; 'if it ain't broke...'
- Dual-stack is the pragmatic answer for ~20 more years
- Some ISPs only do CGNAT — works, but ugly

2.4 IPv6 Address Structure

The phone-number analogy

Example — 2001:db8:cafe:0001 :: a00:27ff:fe9b:1234

Routing prefix /48	Subnet	Interface ID (64 bits)
2001:db8:cafe	0001	a00:27ff:fe9b:1234
≈ Country + Area code	*≈ Neighbourhood*	*≈ Subscriber number*

Compare with a Spanish landline number

+34 91 555 0102

- +34 = country code (Spain) → global routing prefix
- 91 = Madrid area code → subnet ID
- 555 0102 = subscriber → interface ID

2.4 IPv6 Address Scopes

Where this address is allowed to travel

Scope	Range	What it does	Note
Loopback	`::1`	Talk to yourself — always	*Like 127.0.0.1 in v4*
Link-local	`fe80::/10`	Auto-configured per interface; same wire only	*No DHCP needed; always present*
Unique-local	`fc00::/7`	RFC 4193 — internal, not Internet-routable	*v6 equivalent of RFC 1918*
Global	`everything`	Routable on the public Internet	*Your real address*
Multicast	`ff00::/8`	All multicast in v6 — replaces broadcast	*ff02::1 = all hosts on link*

🎯 **KEY TAKEAWAY:** *Every IPv6 NIC is born with at least a link-local fe80:: address — even before any DHCP6 runs.*

2.4 IPv6 Shortening Rules — RFC 5952

Apply once, apply consistently

Rule 1 — Drop leading zeros

in EACH 16-bit group

```
0db8  →  db8
0001  →  1
```

Rule 2 — Replace longest

run of zero groups with :: (only ONCE)

```
0:0:0:0  →  ::
```

(use only once per address)

Worked example

```
Full:        2001:0db8:0000:0000:0000:0000:0000:0001
After R1:    2001:db8:0:0:0:0:0:1
After R2:    2001:db8::1      ✓
```

2.4 EUI-64 — MAC → Interface ID

How SLAAC builds a 64-bit interface ID from a 48-bit MAC

1	Start MAC	`08:00:27:9b:12:34`
2	Split	`08:00:27 | 9b:12:34`
3	Insert FFFE	`08:00:27 FFFE 9b:12:34`
4	Flip U/L bit	`0a:00:27:ff:fe:9b:12:34`
5	Format as IPv6	`0a00:27ff:fe9b:1234`
6	Add fe80:: prefix	**`fe80::a00:27ff:fe9b:1234`**

🎯 **KEY TAKEAWAY:** *Every VirtualBox VM auto-assigns this fe80:: link-local on boot — DHCP not required.*

BLOCK 04

Exercise 2

Project Stage 4 — Address the Network — 1 hour

Exercise 2 — Project Stage 4: Address the Network

1 hour — what your senior admin would actually expect

1

IP plan spreadsheet

Carve 10.42.0.0/22 across 6 VLANs

2

GNS3 router DHCP

ip dhcp pool, default-router, lease, exclusions

3

Static IPs for servers

Ubuntu netplan config, ip addr verify

4

IPFire on VirtualBox

Red/Green/Blue/Orange zones + DHCP server

5

pfSense VLANs + Relay

Inter-VLAN routing, DHCP per VLAN, Option 82

6

Bonus: IPv6 + Wireshark

Dual-stack one interface, capture a fresh DORA

Exercise 2 — Step 1: The Acme SoHo IP plan

Parent block 10.42.0.0/22 → six VLANs, VLSM-allocated

VLAN	Name	CIDR	Hosts	Method
10	Staff	10.42.0.0/25	126	DHCP
20	Servers	10.42.0.128/27	30	Static
30	Wi-Fi BYOD	10.42.0.160/26	62	DHCP
40	Printers	10.42.0.224/28	14	DHCP-reservation
99	Mgmt	10.42.0.240/29	6	Static
100	Voice	10.42.1.0/27	30	DHCP
255	WAN-1	203.0.113.0/30	2	Static

🎯 **KEY TAKEAWAY:** *Always allocate LARGEST FIRST. Always leave 20–30% of the parent free for growth.*

Exercise 2 — Step 2: GNS3 router DHCP for VLAN 10

Cisco IOS — paste, adapt, verify

```
Router(config)# ip dhcp excluded-address 10.42.0.1 10.42.0.20
Router(config)# ip dhcp pool VLAN10_STAFF
Router(dhcp-config)#  network 10.42.0.0 255.255.255.128
Router(dhcp-config)#  default-router 10.42.0.1
Router(dhcp-config)#  dns-server 1.1.1.1 8.8.8.8
Router(dhcp-config)#  lease 0 8     ! 8 hours
Router(dhcp-config)#  domain-name acme.local
Router(dhcp-config)# exit
Router# show ip dhcp binding
```

Verify it works

- Boot a VPCS client on the same VLAN — type 'ip dhcp' — should get an address from the pool
- show ip dhcp binding — proves the lease was committed
- Wireshark on the GNS3 link captures the full DORA exchange

Exercise 2 — Step 4: IPFire on VirtualBox

The four-zone firewall — colour-coded for sanity

RED	GREEN	BLUE	ORANGE
WAN	**LAN**	**Wi-Fi**	**DMZ**
Untrusted Internet — connects via NAT or bridged	Trusted internal network — workstations live here	Wireless guest segment — restricted by default	Public-facing servers — partially trusted

What you do in the lab

- Install IPFire (4 vNICs in VirtualBox), assign each NIC to a zone in setup
- Enable built-in DHCP server for GREEN and BLUE zones (Network → DHCP Server)
- Configure firewall rules: GREEN → RED allow, BLUE → GREEN deny, ORANGE → outside-allow
- Capture screenshots of the DHCP page and the firewall rules table

Exercise 2 — Step 5: pfSense VLANs + DHCP Relay

The senior-admin task

1. **Add VLANs**

   `Interfaces → Assignments → VLANs`

   Create VLAN 10/20/30/40/99/100 on the parent interface; assign each as a new interface.

2. **Per-VLAN DHCP**

   `Services → DHCP Server`

   Enable on VLAN10/30/100; configure pool, gateway, DNS, lease.

3. **DHCP Relay**

   `Services → DHCP Relay`

   On VLAN20, set destination IP = 10.42.0.1 (IPFire). RFC 3046 Option 82 carried.

4. **Inter-VLAN routing**

   `Firewall → Rules → per-VLAN tab`

   Allow VLAN10 → VLAN20 servers ports 80/443/445; default deny everything else.

5. **Verify**

   `Clients on each VLAN`

   Boot a VPCS or VM in each VLAN, run ipconfig /all (Win) or ip addr (Linux). Screenshot each.

Exercise 2 — Deliverables

Project Stage 4 turn-in

Filled-in IP plan

acme_ip_plan_template.csv — every column filled, no missing values

≥ 8 real screenshots

From IPFire and pfSense — DHCP, VLAN config, firewall rules, client verification

Report ≥ 6 pages

Word or PDF — explain WHY each choice, not just what

Bonus: Wireshark capture

.pcapng of a fresh DORA from your lab

Filename

Day04_Project_Stage4_<your_name>.pdf

References — RFCs

The originals — read at least one before next class

RFC	Title	Year
RFC 791	Internet Protocol (IPv4)	1981
RFC 1918	Address Allocation for Private Internets	1996
RFC 1519	Classless Inter-Domain Routing (CIDR)	1993
RFC 2131	Dynamic Host Configuration Protocol	1997
RFC 2132	DHCP Options and BOOTP Vendor Extensions	1997
RFC 3046	DHCP Relay Agent Information Option	2001
RFC 5737	IPv4 Address Blocks Reserved for Documentation	2010
RFC 5952	IPv6 Address Text Representation	2010
RFC 6598	IANA-Reserved Shared Address Space	2012
RFC 8200	Internet Protocol, Version 6 (IPv6)	2017

References — Tools, Standards Bodies, Channels

Where to keep learning

Calculators / Tools

- subnet-calculator.com
- jodies.de/ipcalc
- subnetonline.com
- ipcalc / sipcalc (Linux)
- Wireshark — wireshark.org
- GNS3 — gns3.com
- VirtualBox — virtualbox.org
- IPFire — ipfire.org
- pfSense — pfsense.org

Standards Bodies

- IETF — ietf.org
- IANA — iana.org
- RIPE NCC — ripe.net
- ARIN — arin.net
- ICANN — icann.org
- ITU — itu.int
- IEEE — ieee.org

Recommended Channels

- NetworkChuck
- Practical Networking
- PowerCert Animated Videos
- Sunny Classroom
- Lawrence Systems
- David Bombal
- Ben Eater (binary fundamentals)

End-of-day self-check

If any of these takes more than 30 seconds, re-read the section

1. Convert 192.168.50.0/27 to mask, broadcast, host range — without a calculator.
2. Memorise the four DORA messages and which two are broadcast.
3. Explain why your home router uses PAT and what would break without it.
4. Write the IPv6 link-local that NIC 08:00:27:9b:12:34 auto-assigns.
5. List the three RFC 1918 ranges from memory.

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